

Project Design Phase-II

Technology Stack (Architecture & Stack)

Date	30 September 2026
Team ID	SWTID-2026-2452
Project Name	Import Data using Transform Maps (Spreadsheet)
Maximum Marks	4 Marks

Technical Architecture:

Technical architecture

Spreadsheet > Import Set > Transform Map > Target Table (ServiceNow Personal Developer Instance)

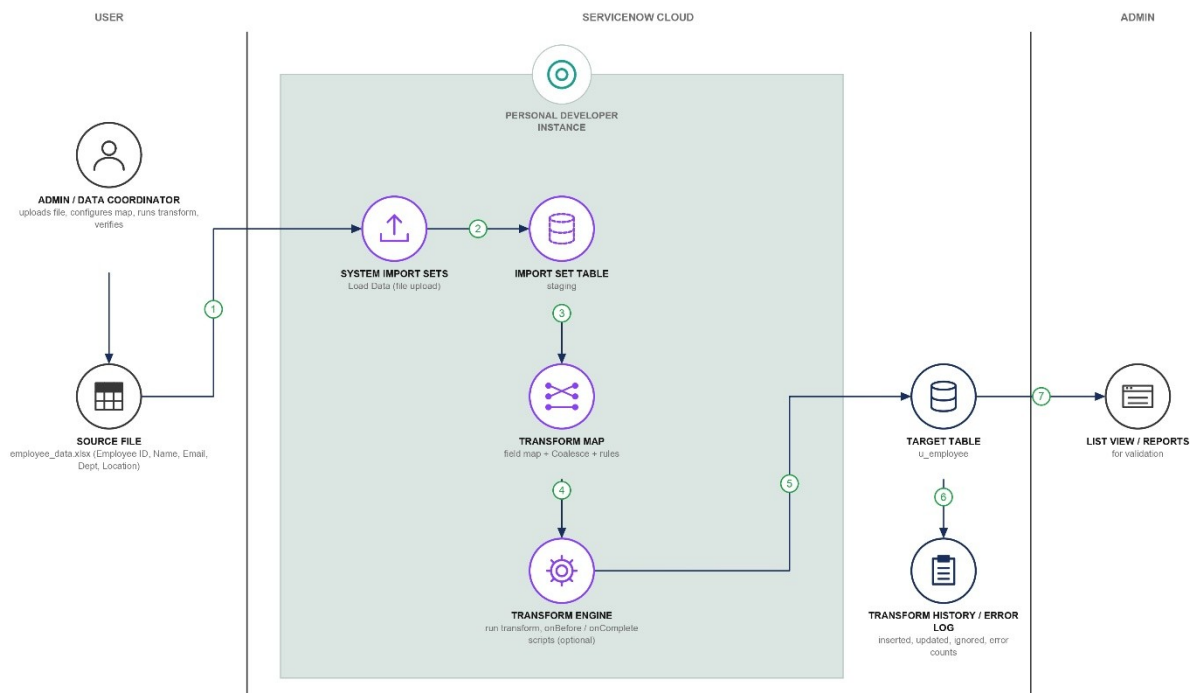


Table-1 : Components & Technologies:

S.No	Component	Description	Technology
1	User Interface	How the administrator interacts with the solution	ServiceNow Platform UI (Next Experience / Classic list & form views)
2	Application Logic-1	Loading the spreadsheet into a staging table	System Import Sets > Load Data (Import Set)
3	Application Logic-2	Mapping source fields to target fields and preventing duplicates	Transform Map, Field Maps, Coalesce
4	Application Logic-3	Optional cleaning / validation logic during transform	Transform Map scripts (onBefore / onAfter, JavaScript)
5	Database	Import Set staging table and target table	ServiceNow Now Platform database – u_import_employee, u_employee
6	Cloud Database	Database service on cloud	ServiceNow cloud instance (Personal Developer Instance)
7	File Storage	Source spreadsheet file	Excel (.xlsx) / CSV file; stored as an attachment during import
8	Infrastructure (Server / Cloud)	Application deployment	ServiceNow cloud (PDI from developer.servicenow.com)

Table-2: Application Characteristics:

S.No	Characteristics	Description	Technology
1	Open-Source Frameworks	No open-source framework is required; source data is prepared in Excel / CSV	Microsoft Excel / CSV
2	Security Implementations	Role-based access to Import Sets and target table; ACLs; encrypted (HTTPS) access to the instance	ServiceNow roles (admin / import_admin), ACLs, HTTPS
3	Scalable Architecture	Same Transform Map can be reused for larger files; platform is multi-tier and cloud hosted	ServiceNow Now Platform
4	Availability	Cloud-hosted instance accessible through a browser	ServiceNow cloud
5	Performance	Bulk load through staging table and batch transform; performance measured by import and transform time	Import Set & Transform engine